VARIABLES IN SPATIAL LEARNING

I. Number of Reinforcements during Training[1]

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Recently Tolman and his co-workers have reported a group of experimental findings (6, 10, 11, 12, 13) which have served as the basis for a more precise formulation of the sign-gestalt theory of learning.

In the initial study (10), rats were subjected to a series of trials on an elevated path to a goal box which contained food. After this preliminary training, the pathway traversed during the trials was blocked and the subjects were permitted to choose one of 12 paths radiating from the point preceding the place at which the original pathway was obstructed. Thirty-six per cent of the rats chose path Number 6 which was the route which led most directly to the original location of the goal box. The responses of the other animals were distributed in a "random fashion" over the other 11 paths. These data were interpreted to indicate "that the disposition which is acquired involves more than the tendency to make the original trained response" (11). They conclude that what was learned in their experimental situation was a cognitive map of that situation, including the *place* of the goal object.

To evaluate the implications of these findings two pertinent questions should be considered. One is how "response-theorists" like Hull can explain the findings of the spatial learning studies. Because of Hull's predilection for working from the relatively simple to the complex, he has concerned himself with the formulation of a behavior theory largely derived from the data of simple learning situations, such as the conditioned response experiments. In such a situation attention is focussed upon a simple response and an attempt is made to ascertain the relationships it bears to the experimental variables. Thus the empirical relationships discovered in such simple experimental situations are to some extent laws of "response reproduction" because the experimenter observes a specific response, such as bar depression, on successive occasions. Consequently the ability of such theories to explain the occurrence of a new response, such as is found in the spatial learning study, has been frequently questioned.

The second problem raised is how those variables known to influence response reproduction are related to spatial learning. If factors underlying response reproduction are unrelated to spatial learning, the theory of behavior based upon conditioning experiments would be seriously questioned. However, if we discover a definite tie-up between the relationships occurring in the two situations

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then it is felt that conditioning theory has implications for our understanding of spatial learning.

The concern of the present paper is with this second question, specifically to ascertain the influence of the number of reinforcements upon spatial learning. Numerous experiments have testified to the importance of this variable in simple learning situations. Perhaps with the aid of the empirical knowledge stemming from such problems as the present, the initial question raised may be more adequately answered. The present experiment in no way attempts to serve as a crucial experiment between conditioning and sign-gestalt theories. At their present stage of development it is questionable whether either theory can generate a precise *a priori* deduction of the results of the present investigation.

EXPERIMENTAL PROCEDURE

Subjects

The subjects for this experiment were 48 male albino rats of the Wistar strain purchased from the General Biological Supply House in Chicago, Illinois. They were approximately 70 days old and had a mean weight of 113 grams at the beginning of the experiment.

Apparatus

The floor plan of the apparatus used is shown in figure 1. Walls and floors were constructed of pine board, $5\frac{1}{2}$ in. by $\frac{3}{4}$ in. so that the width of the maze path was 4 in. The maze was covered with hardware cloth. The doors, indicated by straight, heavy, black lines perpendicular to the walls of the maze, were made of plywood with sponge rubber attached to the bottom of each, making them practically noiseless when manipulated in a vertical direction by strings controlled by the experimenter from behind a screen 59 by 19 inches. Black curtains, indicated by wavy lines were hung 18 inches from the end of each pathway. The entire maze was painted a flat black.

The length of all alleys except the two diagonals to the goal boxes (GB1 and GB2) were 42 inches. The length of the two diagonals were 62 inches.

The construction of this apparatus permitted the animals to be subjected to two separate maze problems. During the training trials, the removable partition, indicated by broken lines in the semi-circular area (see figure 1), was put in place making a typical T-maze problem. The addition of this removable partition plus the distance to the starting box door added 21 inches to the length of the stem of the T-maze, making a total distance of 63 inches to the choice point. During the test trials this partition was removed, the door in the semi-circular area leading to the choice point of the T-maze was lowered, and the animal was permitted to choose one of the six radiating alleys. From the point indicated by a dot in the semi-circular area, these alleys radiated symmetrically at angles of 31, 62, and 90 degrees.

The apparatus was in a room 13 by 18 feet. The maze was placed in a corner of the room with its entrance $2\frac{1}{2}$ feet in front of a table. The experimenter stood behind a screen which was placed at the entrance to the maze. The rats were

transported from the neighboring room in which the living cages were located in a carrying cage with individual compartments. The subjects were kept in these compartments when not being run.

The only light in the experimental room was that provided by a 150 watt reflector-flood bulb placed six feet directly above the choice point of the T-maze.

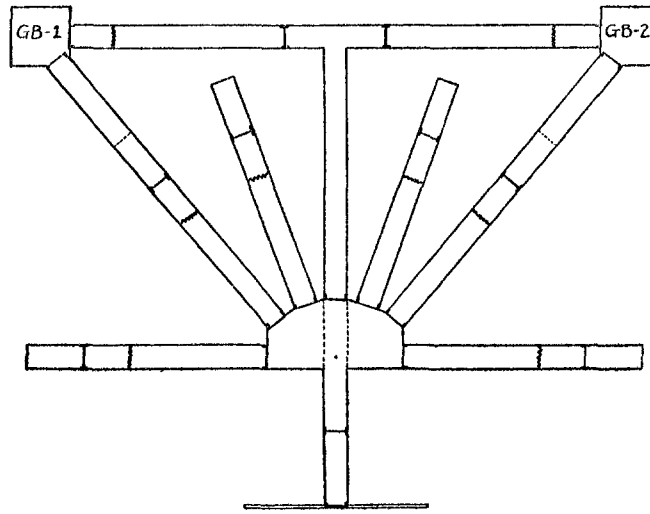


FIGURE 1. FLOOR PLAN OF THE APPARATUS USED

Preliminary Training

The animals were randomly assigned to four groups. Each animal was handled daily for two weeks. One week before the beginning of the experiment the animals were placed on a 24 hour wet-food maintenance schedule. Special feeding cages were utilized and the rats were allowed to remain in these cages for $1\frac{1}{2}$ hours during which time they were permitted to consume a wet mash of Purina Dog Chow kibbled meal. The mean consumption was approximately 10 grams of food, not including water content, per rat per day. Following this $1\frac{1}{2}$ hour feeding period, the animals were returned to their home cages, in which water was always available.

The position habit of each animal was determined on the basis of three non-reinforced trials on the T-maze part of the apparatus.

Training Series

During the training series the animals were subjected to their daily trials 21 hours after the termination of their previous day's feeding.

The T-maze section of the apparatus was used during the training series. Approximately half of the animals were rewarded during the experiment on the side consistent with their position habit, while the other members of the group received their reward on their non-preferred side.

Each subject was given four free choice trials daily. When an animal passed its entire length through one of the raised doors on either side of the choice point, that door was lowered and the animal was permitted to continue to the goal box on that side. If the correct alley was chosen, the rat received $\frac{1}{16}$ teaspoon of a fine mash (Purina Dog Chow checkers which had been dissolved in water) in the goal box. This food was placed on an inverted metal paint-mixing dish whose diameter was 1 and $\frac{1}{8}$ inches. If the incorrect alley was chosen the subject was confined in the empty goal box for a period of 30 seconds.

The subjects had been randomly assigned to four different groups. Each group, during the training series, received a different number of reinforcements to the correct side. A reinforcement was achieved when a subject chose the alley leading to food and consumed the goal object. The identifying numerals of Groups 0, 5, 20,[2] and 100 referred to the total number of reinforcements they received during the training series. When an animal of any group achieved the required number of reinforcements, regardless of which daily trial the criterion was reached, their training series was terminated and on the following day the animal was subjected to the initial trials of the test series.

Test Series

The test trials consisted of four trials daily, in a non-correction situation, for five consecutive days under the same motivational conditions of the training trials, with food being in the same goal box. The partition, in the semi-circular area, as indicated by the broken lines in figure 1, was removed and the door leading to the choice point in the T-maze was lowered while the other six doors in this semi-circular area were raised. Before the first test trial of each subject, the new choice compartment was wiped with a damp cloth to eliminate any possible odor which might have led to tracking. The light was in the same position as it was during the training series.

Whenever an animal passed its entire body length under any of the doors leading from the semi-circular area that door was lowered and the animal's choice recorded. In the recording system used alley 2 represented the 62 degree alley which led directly to the goal box which contained food during the training series. Thus if an animal during the training trials was rewarded in the left goal box (GB1) the six alleys in the test series were numbered one to six from left to right. On the other hand, if the animal had found food in the right goal box (GB2) the alleys were numbered one to six from right to left. Alley 2 always represented the correct choice during the test trials since it represented the 62 degree path leading to the goal box which had contained food during the training trials. Alley number 1 always represented the 90 degree alley next to alley 2.

In order to have all incorrect alleys of the same length, a block was placed in

[2] This experimental group was run by Mrs. Helen C. Mencher whose experiment dovetailed with the present. The experimental training and supervision of both Mr. Gasser and Mrs. Mencher's work was done by the senior author and it is difficult to see how the "experimenter variable" played any role in the results of this investigation.

alley 5, four feet from its entrance. This is indicated in figure 1 by broken lines in both 62 degree alleys.

RESULTS

The results of the test trials are analyzed in terms of the performance exhibited by the various groups on the first test trial as well as the entire test series.

Table 1 indicates the results of the first test trial. For the purpose of clarity it may be useful to repeat that the numerals assigned to the six alleys of the test series have as their point of reference the goal box which contained the food during the training series. The 62 degree alley which led to this goal box is referred to as alley 2. In case this goal box was on the left side during the training trials then the alleys were numbered from one to six in a clockwise fashion. On the other hand if the correct goal box was on the right side during the training series, the alleys were numbered one to six in a counter-clockwise fashion.

TABLE 1

Showing the number and per cent of each group responding to each of the six alleys on the first test trial

GROUP	N	ALLEY 1		ALLEY 2		ALLEY 3		ALLEY 4		ALLEY 5		ALLEY 6	
		N	%	N	%	N	%	N	%	N	%	N	%
0	12	6	50.0	0	0.0	1	8.3	0	0.0	1	8.3	4	33.3
5	11	4	36.4	1	9.1	2	18.2	0	0.0	1	9.1	3	27.3
20	12	5	41.7	1	8.3	1	8.3	2	16.7	0	0.0	3	25.0
100	13	7	53.9	0	0.0	3	23.1	0	0.0	1	7.7	2	15.4

The data of table 1 indicate that only 2 of 36 animals (5.56 per cent) that had training on the T-maze responded to the new situation by choosing alley 2, while the most frequent response in each group, for the first test trial, was to alley 1.

Table 2 shows the mean number of responses exhibited by each group to each alley during the twenty test trials, while table 3 evaluates the significance of the differences between the selection of the correct path (alley 2) by the various groups during the test series.

Although the response to alley 2 was the only one rewarded during the test series, the subjects in each group chose the two 90 degree alleys (alleys 1 and 6) more frequently than they chose the rewarded path. In spite of this, there was significant difference between the frequencies with which the response to alley 2 was exhibited (see table 3) by the different experimental groups and this fact suggest that number of reinforcements is a pertinent variable in spatial learning.

Analysis of variance type of statistical treatment was precluded because of the unequal variances. It is evident that the distributions of responses to alley 2 are skewed since the lowest possible number of correct choices, zero, is less that 1.8 sigma units from the mean in all groups. A technique devised by Festinger (1) which is appropriate in testing the means from skewed distributions was used to

obtain the significances of the differences shown in table 3. Not only do the data of table 3 suggest that number of reinforcements influences the degree of spatial learning but also indicates the nature of such a relationship, i.e. spatial learning increases with increasing number of reinforcements in the present situation from 0 to 20 and then decreases somewhere between 20 and 100 reinforcements.

TABLE 2

Showing the mean number of responses exhibited by each group to each of the six alleys during the 20 test trials

GROUP	ALLEY 1		ALLEY 2 CORRECT ALLEY		ALLEY 3		ALLEY 4		ALLEY 5		ALLEY 6	
	M	σ	M	σ	M	σ	M	σ	M	σ	M	σ
0	8.08	3.43	2.50	1.50	1.33	.94	2.00	1.08	1.75	.92	4.33	1.70
5	5.73	1.76	3.64	2.50	2.46	1.16	2.09	1.00	1.55	1.16	4.36	1.67
20	6.33	4.59	5.08	3.35	1.92	1.04	1.17	.99	1.92	1.26	3.58	2.06
100	6.77	4.92	3.31	2.30	2.77	1.31	1.39	1.08	1.31	.99	4.46	3.43

TABLE 3

Comparison of the mean number of correct responses exhibited by the different groups during the 20 test trials

COMPARISON	MEANS	d.f.	F	LEVEL OF SIGNIFICANCE
20/0	5.083/2.500	72/67	2.03	1%
20/100	5.083/3.308	72/54	1.54	5%
20/5	5.083/3.636	72/69	1.40	20%
5/0	3.636/2.500	69/67	1.45	20%
5/100	3.636/3.308	69/54	1.10	—
100/0	3.308/2.500	54/67	1.32	20%

DISCUSSION OF RESULTS

The data of our initial test trial is at variance with the results obtained by Tolman, Ritchie and Kalish (10). They discovered that 36 per cent of their animals chose the "path which pointed directly toward the location of the goal" while our results reveal that only 5.6 per cent behaved in a similar manner. Since the present experiment was not intended to be a repetition of their experiment, the differences obtained can probably be ascribed to one or several of the important differences between our respective experimental techniques. However, such variations were not merely a result of chance.

Tolman has repeatedly attempted to delineate between theories which assume that "responses" are learned and his theory which states that a cognitive map is acquired. Obviously the definition of the term "response" is of utmost importance and one must be careful not to restrict the meaning of such a term to only spatial responses, i.e. turning left or right. The concern of the conditioning theorist for problems in the field of discrimination learning testifies to the fact

that approach responses to a stimulus can be acquired. Since rats can learn to make a response *towards* a light or other types of stimulus patterns, there appears to be a decided advantage in the use of walled mazes as compared with elevated mazes in investigating the theoretical implications of spatial learning studies.

Because rats according to the "response theorist" can acquire responses toward stimuli it is important to separate experimentally such responses from behavior which may be interpreted as evidence for "pure" spatial learning. The light in the present experiment therefore was placed directly above the choice point in the T-maze. During the training series the rats in this experiment were restricted by inserting partitions in the semi-circular area which served as a choice point in the test trials. Therefore the possibility of our animals acquiring certain approach responses to specific aspects of the extra-maze environment, aside from the light was limited when compared to the situation confronting the animals in the study by Tolman, Ritchie and Kalish (10). These authors do not assert that spatial learning can occur in an experimental situation void of all stimulus cues but do imply that such cues are not responded to directly, but rather serve as signposts or landmarks by which the animal may orient itself. The ideal experimental environment for a spatial learning experiment would seem to be one in which only one definite stimulus cue would be present, at a position other than the goal box.

In attempting to ascertain the factors relevant to the choices exhibited by the experimental groups for the first test trial, one must not only explain the preference exhibited for alley 1, which included the same turning response rewarded during the training trials, but must also consider the large number of choices to alley 6, which involved a response opposite to the one initially reinforced. The combined per cent of responses to alley 1 (44.4 per cent exhibited by Groups 5, 20, and 100 on the first test trial) is significantly greater (at the 4 per cent level of confidence) than their per cent of choices to alley 6 (22.2 per cent). This, plus the fact that the relative preference for alley 1 over alley 6 increased with increasing reinforcements from 5 to 100 suggest that the tendency to make a 90 degree response in the direction rewarded was the strongest turning response tendency acquired during the training series. In spite of the fact that responses to alleys 2, 3, 4, and 5 were more similar in their pattern of proprioceptive stimuli to the response to alley 1 than was the response to alley 6, this latter response was the second most preferred. The most likely explanation of this phenomenon is that the new environment induced by the test trial instigated emotional responses in the subject resulting in their going to the darkest part of the maze. Since the only light source was located above the choice point of the T-maze, alleys 1 and 6 were the most shaded.

Regardless of the results of the first test trial the data of the 20 test trials suggest that the number of reinforcements is one variable determining the degree of spatial learning. Such a relationship must be accounted for by the learning theorist. For a sign-gestalt theorist it may be necessary to distinguish between the relationship, number of reinforcements bears to the strength or "clarity" of a specific sign-gestalt-expectation and a group of sign-gestalt-expectations (cogni-

tive map). Probably the sign-gestalt-expectation for food in the rewarding goal box in the T-maze was "clearer" for group 100 than for group 20 but the cognitive map of the larger environmental situation was "clearer" for group 20.

For a conditioning theorist it appears essential that more of the laws of response generalization and the patterning of simulus compounds be ascertained. These are complex problems and certainly complexity should not serve as an escape clause for any theory. But one should be aware of the mechanisms by which a specific theory may be able to handle certain phenomena. In the training series of the present experiment (as well as in the preliminary trials of the experiment by Tolman, Ritchie, and Kalish (10)) it should be realized that not only were many S-R associations being formed and strengthened, but also that several responses were being connected to the same or similar stimlus situations. Certain turning responses, approach responses, forward going responses, etc., occurring at specific parts of the maze were all being followed by reward and consequently their connection to the instigating stimulus situation received increments in habit strength. The principles governing the interaction of these various associations at the presentation of a relatively new stimulus compound such as was offered in the test trials, hold the "key" to the problem of spatial learning for the conditioning theorist. Unfortunately the data of the present investigation merely suggest the theoretical problem to be considered, rather than contribute to its solution. Perhaps under conditions of overlearning, turning responses achieve stronger habit loadings than do approach responses. It should be emphasized, however, that there is nothing inherent in a behavior theory based upon conditioning principles, which precludes the prediction of a response not previously evoked. It has been repeatedly demonstrated that with the use of certain theoretical mechanisms, that the conditioning theorist has been able to predict the appearance of novel responses (2, 5, 7).

The problem of what is learned, with which Tolman and his co-workers concern themselves, needs to be subjected to careful analysis so that we may be sure that the specific area of controversy be known. According to the type of theory construction which Tolman (9) has been so instrumental in bringing to the attention of the learning theorist, a group of theoretical variables intervene between the independent and dependent variables. The only meaning such intervening variables have in theories such as those of Hull (2) and Tolman (8) are the relationships they bear to objectively defined variables. The pragmatic justification for the utilization of such intervening variables is that they generate theoretical estimates of the response in agreement with the observed dependent variable. All theories therefore must lead to estimates of *responses*, and therefore in this sense, must be "response theories." The real problem is not what is learned, but rather at what step in our theorizing do we begin to introduce the theoretical estimate of the response. The construct habit strength ($S^{H}R$) appears early in Hull's attempt to bridge the gap from the independent to the dependent variables (p. 383, 2). Although this construct does not yield a precise quantitative estimate of the response, at this stage, it does specify its nature i.e., right turning or bar depressing, etc. It would seem that in Tolman's theory

that the nature of the to-be-estimated response appears at a later stage. In fact one criticism which may be justifiably leveled against the sign-gestalt theory is that it has never been very explicit as to how a cognitive map is translated into specific responses. Perhaps when this is done, this area of apparent disagreement of what is learned, may disappear.

SUMMARY

The purpose of the present investigation was to determine the influence of the number of reinforcements, in a free choice T-maze situation, upon the amount of spatial learning. Forty-eight male albino rats were used as subjects.

The problem confronting the subjects during the preliminary training was to choose the side leading to food in an enclosed T-maze, with a 150 watt diffused light above the choice point. The population was divided into four groups, receiving respectively 0, 5, 20, and 100 reinforcements following their correct responses. During the 20 test trials the route to the choice point was blocked, and the animals had a choice of one of six possible pathways radiating symmetrically at angles of 31, 62, and 90 degrees from the new choice point. One of the 62 degree pathways led directly to the goal box which contained the reward during the preliminary training, and the response to this pathway was the only one rewarded during the test series.

In the initial test trial only two animals (one each in Groups 5 and 20) chose the alley leading directly to the food while the modal response was to the "90 degree" alley which coincided with the right or left turning response which had been rewarded during the training trials. During the 20 trials of the test series, Groups 0, 5, 20, and 100 had means of 2.50, 3.64, 5.08, and 3.31 correct responses, respectively. One of these differences was significant at the 1 per cent level, one (Groups 20 and 100) at the 5 per cent level, and three were significant at the 20 per cent level.

The results indicate (1) that varying number of reinforcements has a differential effect upon spatial learning, and (2) that spatial learning increases with greater number of reinforcements up to 20 but decreases at 100 reinforcements. The theoretical implications of these results are discussed.

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